Yihao Wang

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EDUCATION AND RESEARCH APPOINTMENT

Peking University Shenzhen Graduate School | Shenzhen, China 09/2026 – Present
Research Assistant

University of Southern California | Los Angeles, CA 01/2025 – 08/2026
Master of Science in Applied Data Science; graduated August 25, 2026

Hong Kong Baptist University | Zhuhai, China 09/2020 – 07/2024
Bachelor of Science (Honours) in Computer Science and Technology, GPA: Overall 3.46/4.0 , IELTS: 7

PUBLICATIONS AND MANUSCRIPTS

Yihao Wang et al. (2026). "SAGE: A Statistical Acceptance Gate for Self-Evolving Agents". *AAAI 2027, under review (Phase 2).*

Linhan Xia, Rui Liu, Zhaofeng Zhang, **Yihao Wang**, Shengxin Zhu. (2026). "LoCA: Forward-Only LLM Tuning after One-Shot Calibration with Local Credit Assignment". *AAAI 2027, under review (Phase 2).*

Jiaqi Li, Qi Pan, **Yihao Wang**. (Mar. 2024). "Sentiment analysis applied on Amazon reviews". *International Conference on Machine Learning and Automation*. *EI index*, (Online).

Jianxin Ye, Zeqiao Huang, **Yihao Wang**, Jinming Chen. (Dec. 2023). "Fine Grain Classification of Dog Breeds Based on HERBS Method". *2023 IEEE International Conference on Electrical, Automation and Computer Engineering* (ICEACE). *EI index*, (Online).

RESEARCH EXPERIENCE

Network Root-Cause Analysis | Peking University 09/2026 – Present

- Framed root-cause analysis as a causal claim that must be verified before acting: counters, voting and LLM agents output a label with nothing to test, and blind actions such as closing a lossy link can turn loss into congestion.
- Built a world model with an analytic skeleton (water-filling, DCTCP, hop-by-hop loss) where each fault enters through an explicit term, plus a Transformer residual for what the skeleton misses; trained on paired ns-3 twin runs with shared seeds.
- Designed a propose–verify–abstain–undo chain: an amortized head nominates 129 candidate causes incl. "no fault", replay likelihood scores each, the system abstains when top causes are close, and undoing the cause in the model predicts recovery.

SAVE: Verified Self-Distillation for LLM Agents | Peking University 06/2026 – Present

- Identified the self-sampling bottleneck of STaR-style self-distillation: inputs the model never solves are discarded by rejection sampling, so training can only reinforce behaviors the current model already reaches.
- Developed SAVE: an agent distills failure patterns (wrong function, bad arguments, malformed output) into a temporary harness; regenerations kept only if a deterministic verifier accepts them are distilled, scaffold-free, with the original successes.
- Ran five rounds of function-calling self-distillation under matched data, token and step budgets: scaffold-free accuracy 95.9% (4,896/5,106) vs. 71.9% (3,673/5,106) with self-sampled successes only; a round-1 near tie widened to 24.0 points.

SAGE: Statistical Acceptance Gate for Self-Evolving Agents | Peking University 05/2026 – Present

- Identified the acceptance step as the overlooked bottleneck of skill self-evolution: keeping any edit that raises the aggregate score admits permanent regressions and commits lucky edits (Optimizer's Curse); naive-gate regression rates reach 45.6%.
- Proposed SAGE: per-item paired evaluation with an asymmetric regression cost $\lambda \geq 1$ and a regression cap, plus a one-sided exact binomial test on discordant pairs (exact McNemar at $\lambda = 1$); it abstains otherwise and recovers the SkillOpt gate at (1, 1, 1).
- Equal-budget evaluation (SkillOpt optimizer fixed) on 4 backbones × 5 benchmarks: regression rate lower in 19/20 settings, zero in 13 (LiveMath 36.5%→0%, OfficeQA 42.8%→0%); best final score in all 20, +8.73 points over SkillOpt on average (paired comparison +6.20, test +2.53).

PROFESSIONAL EXPERIENCE

Henan Cdcenter Network Technology Co., Ltd. | Backend Engineering Intern, Zhengzhou 06/2024 – 12/2024

- Maintained Hadoop-based data warehouses and developed Python ETL scripts for relational and unstructured data; used Hive/Spark for distributed processing to support business reporting and real-time analytics.

SKILLS

- **LLMs & Agents:** Hugging Face Transformers, LoRA / QLoRA fine-tuning, RAG, LangChain
- **Programming:** Python, SQL, R, C
- **Machine Learning:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, SVM, PCA, GAM
- **Data Engineering:** Spark, Hive, Hadoop, Python ETL
- **Research Tools:** Git, NumPy, pandas, SciPy, matplotlib, Jupyter, LaTeX